Project: Clean up Guides and Working with Orphaned Users and Orphan file Residues:

Concept: Admin Work with File Cleanings and Decluttering

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Things to Consider:

Before we start putting in the commands, it is good to think over couple of questions. What does it even mean the Orphaned Users? How do they look like in real Systems? What does it even mean to have Orphaned Files? These questions unlock the answers and perspective to work with them. My understanding is simple, Orphaned Users create Orphaned files residue, and if they are not cleared, they cause great deal of security risk to the system. Let's dive into it in a very simple way, even a 10-year-old would understand.

1. What is Orphaned User?

   As the term, "Orphan" is self-explanatory. In Systems, when real people leave the company or get transferred to the different departments, their older accounts, for some reason, are still active, or have access privileges to their previous department. Those accounts are called Orphaned Users.

   Orphaned Users are of two types: System users and Normal Users.

   i. System users: for eg: root, daemon, nginx, www-data, systemd-resolve and many more.
   ii. Normal users: alex001, bob_cool05, nancy555, and many more.

2. What about Orphaned files Residue?

   When these users (both Normal or System) when they are no longer active, meaning (normal users deleted, or running services are no longer active), their files still might be lingering on the machines. These files do not have any real owner to them except for their older owner's UUID. These files are Orphaned Files.

3. What Security Risk, these orphaned users and Files do possess?

Without any Fancy Technical Jargons, I'll keep it Simple.

Scenario1:

In this scenario: Bob works in Accounting Department, and his job responsibilities include working with customers' sensitive data including credit cards information. For this he needs certain level of 'access rights'. One day Bob gets fired and leaves the company, and let's suppose, IT department forgets to delete Bob's useraccount. Now, his account is orphaned user. And if somebody gets hold of the bob's account, the customers data would be at great risk. Or, If Bob wants to take revenge, he can also do that, because he has top level of access rights assigned to his user account.

The same logic applies to the System users and Orphaned Files. Though it is little different, older services might have some files, or older system users might have certain level of access-rights. Similarly, Orphaned files also might contain sensitive information for the company, which could be misused, if it gets to the wrong hands. In this way, the orphaned users threaten the Company's Information Security.

Now the real guides:

1. System keeps users list in /etc/passwd file
2. System keeps users' encrypted passwords in /etc/shadow file.

The trick is to find active users on the System.

1. Check if some users' information is not synced in these files (userA might be in /etc/passwd but not in /etc/shadow file and vice versa). If some found, that is orphaned user.
2. If checking system users, list the active current system users and check other listed system users if they own any files. If some users do not own any files, they might be potentially orphaned users.
3. Working with the files, *sudo find / -nouser -o -nogroup 2>/dev/null* will find the list of files not owned by any real user. These files will only show previous UUID as their owner. If we see numbers instead of useraccount, know it is orphaned file.

In this way, I have simplified the task of Identifying and removing Orphan users. I have also demystified the concept of whole orphan users and orphaned file residue. With this guide, anyone can google the command and use it for their purpose. If you need more clarification, please check out my other hand drawn figure, that will add more clarification to the topic.